Very good:: year-by-year roadmap for a full-time PhD in Computer Science at IIT Bombay

10 messages

Gautham <gauthamupi4@gmail.com> Sat, 27 Dec, 2025 at 13:45
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Here's a tailored year-by-year roadmap for a full-time PhD in Computer Science at IIT Bombay, based on the department's structure (CSE) and general institute rules. It typically spans 4-5 years, emphasizing coursework, qualifiers, research publications, and thesis defense. Timelines can vary by supervisor and progress, with annual progress seminars (APS) and reports (APR) mandatory each year.

Year 1: Coursework and Qualifiers

- **Semester 1-2**: Complete 16-32 credits (minimum for MTech entrants), including core CS courses (e.g., Advanced Algorithms, Machine Learning, Theory of Computation), electives, and a seminar. Enroll in non-credit requirements like research methodology and teaching assistantship.
- **By End of Year**: Pass the comprehensive/qualifier exam (written + oral on coursework/research aptitude). Submit and present initial research proposal to a committee. Identify thesis topic and supervisor.
- **Milestones**: 1-2 seminar presentations; secure funding (e.g., institute fellowship via GATE).

Year 2: Research Ramp-Up

- **Focus**: Dive into literature review, experiments, and prototype development (e.g., AI models, systems, or algorithms). Attend workshops/conferences.
- **Key Tasks**: Submit first APR; deliver APS on preliminary results. Aim for 1 conference paper submission (e.g., NeurIPS, ICML affiliates).
- **Milestones**: Confirm PhD registration post-qualifiers; form thesis committee.

Year 3: Core Research and Publications

- **Focus**: Execute main experiments, data analysis, and iterations. Collaborate on interdisciplinary projects if applicable.
- **Key Tasks**: Second APR and APS; target 2-3 publications in top venues (e.g., ACM, IEEE journals/conferences). Draft thesis chapters 1-3 (introduction, literature, methods).
- **Milestones**: Mid-program review; demonstrate substantial progress toward novel contributions.

Year 4: Thesis Synthesis and Defense

- **Focus**: Compile results, write full thesis, and respond to feedback.
- **Key Tasks**: Third APR/APS; submit thesis to external examiners (2-3 months review). Prepare open seminar and viva voce.
- **Milestones**: Defend thesis successfully; incorporate revisions for degree award (typically by mid-year).

Year 5 (If Extended): Finalization

- **Focus**: Address examiner comments, publication polishing, or minor extensions.
- **Key Tasks**: Final submissions; maximum registration is 7 years, but CSE averages 4.5 years.
- **Milestones**: Degree conferral; job/postdoc applications.

This roadmap aligns with IIT Bombay's PhD rules (e.g., credits within 24 months, thesis within 7 years) and CSE-specific expectations for publications (3-5 by defense). Track progress via the academic portal.

Would you like me to adjust this for a specific CS sub-area like AI/ML or add details on funding/stipends?

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Year 2 and Year 3 in IIT Bombay's full-time Computer Science PhD mark a shift from foundational research setup to intensive output and validation, with Year 2 focusing on exploration and Year 3 on consolidation and external validation.[2][3]

Key Differences
Aspect	Year 2 Focus	Year 3 Focus
Primary Activities	Literature review, prototype development, initial experiments (e.g., algorithm testing or model training).	Major experiments, data analysis, iterations, and thesis framework drafting.
Publications	Aim for 1 conference paper submission (e.g., early workshops).	Target 2-3 publications in top venues (e.g., NeurIPS, ICML, ACM).
Reviews & Seminars	First Annual Progress Seminar (APS) and Report (APR) on preliminary results; post-qualifier registration confirmation.	Second APS/APR; mid-program committee review for substantial progress.
Depth of Work	Building skills and supervisor alignment; attend conferences.	Novel contributions, collaborations, and chapters 1-3 (intro, literature, methods).
Risk Level	Lower; flexibility to pivot topics.	Higher; must demonstrate viability for defense trajectory. [2][3]

Transition Impact
Year 2 builds momentum post-coursework, emphasizing independence under guidance, while Year 3 ramps up accountability with publication pressure and progress gates, often determining funding continuation or extensions.[3]

[1](https://www.reddit.com/r/GATEtard/comments/1kr2rue/got_offers_in_iit_bombay_cminds_ms_and_iisc_mtech/)
[2](https://www.youtube.com/watch?v=QGvTIBXlVQk)
[3](https://www.iitb.ac.in/newacadhome/phdRules.pdf)
[4](https://www.reddit.com/r/developersIndia/comments/n86ldy/3_years_of_experience_or_2_years_of_mtech_from/)
[5](https://www.cse.iitb.ac.in/admissions/pg)
[6](https://www.ieor.iitb.ac.in/faq/phd-admissions)
[7](https://www.facebook.com/groups/1015191011986167/posts/1761035704068357/)
[8](https://www.cse.iitb.ac.in/admissions/pg/ms)
[9](https://iitsystem.ac.in/sites/default/files/externalpeer-review/2/9a858dbac3e6e3cd871e7c676.pdf)
[10](https://www.iitb.ac.in/newacadhome/Ph.D._Brochure_2024-25.pdf)
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Literature review, prototype development, and initial experiments form the core of Year 2 research initiation in IIT Bombay's Computer Science PhD, building a strong foundation post-qualifiers.[2][5]

Literature Review
Conduct a comprehensive, critical survey of 50-100+ papers from top venues (e.g., ACM, IEEE, NeurIPS) using tools like Google Scholar or DBLP. Identify gaps, synthesize trends (e.g., in AI scalability or algorithm efficiency), and draft a 20-30 page chapter with analytical comparisons, not just summaries. Present findings in the first Annual Progress Seminar (APS) for feedback.[4][5][2]

Prototype Development
Implement a basic proof-of-concept system, such as a novel ML model in PyTorch/TensorFlow or an optimized algorithm in C++/Python. Focus on feasibility testing with synthetic/small datasets; iterate based on supervisor input. Document architecture, code, and early metrics (e.g., runtime, accuracy) for the Annual Progress Report (APR).[11]

Initial Experiments
Run preliminary tests like ablation studies, benchmarks against baselines (e.g., comparing your algo to SOTA on UCI datasets), or simulations. Collect initial data/results to validate hypothesis; aim for one workshop/conference submission. These feed into APS, ensuring topic viability before deeper commitment.[6][11]

[1](https://iitsystem.ac.in/sites/default/files/externalpeer-review/2/9a858dbac3e6e3cd871e7c676.pdf)
[2](https://www.cps.iitb.ac.in/admissions/doctoral-programme/)
[3](https://www.zenithphd.com/phd-in-mangment-at-iit-bombay-complete-guide/)
[4](https://www.youtube.com/watch?v=ZwUZZHYNYiE)
[5](https://acad.iitb.ac.in/files/2024-04/Dissertation17june09-10.pdf)
[6](https://www.ee.iitb.ac.in/course/~hemantr/details/PhDHandbook-07-08.pdf)
[7](https://www.iitb.ac.in/newacadhome/Ph.D.Brochure2023-24.pdf)
[8](https://www.scribd.com/document/713141841/Iit-Bombay-Phd-Thesis-Template)
[9](https://acad.iitb.ac.in/admissions/research)
[10](https://www.reddit.com/r/Indian_Academia/comments/1bur76n/any_tips_for_the_statement_of_purpose_for_phd_at/)
[11](https://www.iitb.ac.in/newacadhome/phdRules.pdf)

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Year 3 in IIT Bombay's Computer Science PhD shifts to core research execution and validation, building on Year 2's foundations with greater emphasis on rigor, output, and trajectory toward defense.[1]

Major Experiments
Execute large-scale experiments, such as full dataset training (e.g., on ImageNet or custom CS benchmarks), hyperparameter tuning, and ablation studies to test hypotheses. Address scalability issues in algorithms or models, iterating based on failures to generate robust results.[11][1]

Data Analysis and Iterations
Analyze results quantitatively (e.g., statistical significance, error bounds) and qualitatively, refining methods through multiple cycles. Draft thesis chapters 1-3: introduction, related work synthesis, and methodology with preliminary findings.[1]

Key Milestones
Deliver second Annual Progress Seminar (APS) and Report (APR); undergo mid-program committee review to confirm "finish line" visibility. Target 2-3 publications in top venues (e.g., NeurIPS, ICML, ACM SIGPLAN) and explore internships/post-PhD plans.[11][1]

[1](https://www.cse.iitb.ac.in/~sri/talks/Guiding-PhD-students-Feb2023.pdf)
[2](https://acad.iitb.ac.in/files/MTECH_BROUCHURE2025-26.pdf)
[3](https://www.instagram.com/p/C6sHcCrI7jc/)
[4](https://www.youtube.com/watch?v=oX3xhE1EDzU)
[5](https://www.cse.iitb.ac.in/admissions/pg/phd)
[6](https://www.ee.iitb.ac.in/web/phd-admission/)
[7](http://www.iitb.ac.in/newacadhome/phdbrochure201315Marchnew.pdf)
[8](https://www.facebook.com/PhDsofIndia/photos/i-did-my-phd-from-iit-bombay-in-a-relatively-new-department-called-educational-t/880465730762587/)
[9](https://timesofindia.indiatimes.com/city/mumbai/iit-bombay-first-in-india-to-award-more-than-400-phds-in-a-year/articleshow/93684470.cms)
[10](https://www.iitb.ac.in/newacadhome/Ph.D._Brochure_2024-25.pdf)
[11](https://www.iitb.ac.in/newacadhome/phdRules.pdf)
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Year 2 and Year 3 in IIT Bombay's Computer Science PhD differ in depth, scale, and output expectations, with Year 2 emphasizing exploration and setup, while Year 3 demands rigorous validation and publication momentum.[11][12]

Key Differences with Example (AI Model Optimization)
Aspect	Year 2 (Exploration Phase)	Year 3 (Validation Phase)
Experiments	Initial tests: Train a basic neural net on small dataset (e.g., MNIST subset) to test hypothesis on edge optimization. Quick iterations, synthetic data.	Large-scale: Full training on ImageNet/CIFAR-100 with distributed GPUs; ablation studies across 10+ architectures, real-world benchmarks.
Analysis Depth	Preliminary metrics (e.g., 85% accuracy baseline vs. simple tweak). Identify issues like overfitting.	Statistical rigor: t-tests, confidence intervals, error analysis; compare vs. 5+ SOTA models (e.g., ResNet vs. your variant).
Outputs	1 workshop paper draft; prototype code on GitHub.	2-3 conference papers (e.g., ICML submission); draft thesis chapters with results tables/figures.
Reviews	First APS: Show feasibility ("This approach works in toy settings"). Pivot if needed.	Second APS/mid-review: Prove novelty ("10% speedup over SOTA with proofs"). Funding gates tighten. [11][1]

Example Transition Impact
In Year 2, you prototype a custom attention mechanism yielding preliminary gains on toy NLP tasks. Year 3 scales it to BERT-scale models, publishes validated results, and pivots to deployment challenges, setting defense trajectory.[11]

[1](https://www.youtube.com/watch?v=QGvTIBXlVQk)
[2](https://collegedunia.com/university/25703-iit-bombay-indian-institute-of-technology-iitb-mumbai/phd-computer-science-and-engineering-2691)
[3](https://iitsystem.ac.in/sites/default/files/externalpeer-review/2/9a858dbac3e6e3cd871e7c676.pdf)
[4](https://www.cse.iitb.ac.in/admissions/pg)
[5](https://www.ieor.iitb.ac.in/faq/phd-admissions)
[6](https://www.facebook.com/groups/1015191011986167/posts/1761035704068357/)
[7](https://www.iitb.ac.in/newacadhome/Ph.D.Brochure2023-24.pdf)
[8](https://www.reddit.com/r/Btechtards/comments/1hsev3t/ama_phd_from_old_iit_doing_phd_in_computer/)
[9](https://www.cse.iitb.ac.in/admissions/pg/ms)

[10](https://acad.iitb.ac.in/admissions/research)
[11](https://www.cse.iitb.ac.in/~sri/talks/Guiding-PhD-students-Feb2023.pdf)
[12](https://www.iitb.ac.in/newacadhome/phdRules.pdf)
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IIT Bombay's MS by Research program in Computer Science emphasizes research from the start, differing from typical MTech structures that focus more on coursework. Year 1 involves completing coursework, R&D projects, and qualifiers, with specific milestones for progression.

Coursework Requirements
Students must complete a minimum of 58 course credits plus 6 credits for Communication Skills, typically spread over the first two semesters. This includes at least 38 elective credits (minimum 12 per semester), core courses where applicable, a 4-credit seminar in Semester I or II, and 16 credits of MS R&D projects (8 credits each in Semesters I and II). Non-credit elements like teaching assistantships align with the described MTech path but are integrated differently in MS.[1]

Qualifiers and Exams
No explicit "comprehensive/qualifier exam" is mandated like in some MTech programs; instead, progression relies on coursework performance, R&D projects, and supervisor evaluation by the end of Year 1. Students identify a thesis topic and supervisor early, with seminar presentations serving as research aptitude checks.[1]

Key Milestones
- 1-2 seminar presentations and Communication Skills course clearance within first two semesters.
- Secure funding via institute fellowships (often GATE-based for eligible entrants).
- By end of Year 1: Complete initial R&D projects; transition to MS Thesis (Semesters 4-6 possible).[1]

Comparison to MTech
Aspect	MTech (as described)	MS Bombay [1]
Credits Year 1	16-32 coursework + non-credits	58 coursework + 6 comm skills + 16 R&D
Qualifier	Written/oral exam + research proposal	R&D projects + seminar evaluation
Research Focus	Initial proposal by Year 1 end	R&D from Semester 1; thesis Sem 4+
Funding/TA	GATE fellowship + TAship	Similar, but research-oriented

MS Bombay suits research careers with lighter initial coursework but mandatory early R&D, contrasting MTech's heavier taught components before qualifiers.[1]

[1](https://acad.iitb.ac.in/files/MS%20by%20Research%20programme.pdf)
[2](https://lsu.edu/eng/cse/programs/graduate_program/files/mastersplanofstudy.pdf)
[3](https://acad.iitb.ac.in/admissions/masters/msbyresearch)
[4](https://www.lsu.edu/eng/cse/programs/undergraduate_program/undergraduate_major.php)
[5](https://www.cps.iitb.ac.in/admissions/masters-programme/)
[6](https://computing.louisiana.edu/cacs-research/academic-programs/program-prerequisites)
[7](https://acad.iitb.ac.in/admissions/masters/msresfaq)
[8](https://catalog.latech.edu/content.php?catoid=25&navoid=1244&filter%5Bitem_type%5D=3&filter%5Bonly_active%5D=1&filter%5B3%5D=1&filter%5Bcpage%5D=14&print=&coid%5B0%5D=139576&display_location=filter_template&contract)
[9](https://www.geeksforgeeks.org/gfg-academy/iit-bombay-through-gate/)
[10](https://latech.catalog.acalog.com/content.php?catoid=7&navoid=289&filter%5Bitem_type%5D=3&filter%5Bonly_active%5D=1&filter%5B3%5D=1&filter%5Bcpage%5D=13&print=&expand=1)
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IIT Bombay's MS by Research program serves as a research-focused stepping stone, often leading to PhD, while the PhD demands deeper, longer-term original contributions.[1][3]

Duration and Completion
MS by Research typically spans 1.5-3 years, with early exit possible for exceptional research output like publications. PhD requires 4-6 years minimum, emphasizing a comprehensive thesis with sustained novelty, coursework (around 16-24 credits), and qualifying exams.[3][1]

Research Emphasis
MS prioritizes R&D projects from Semester 1 (16 credits), seminars, and electives tailored to research (58+ credits total), aiming for industry R&D or PhD prep. PhD builds on this with full-time thesis work, annual progress reviews, and publications in top venues, without fixed credit caps for research.[2][3]

Key Differences

Aspect	MS by Research [3]	PhD [1]
Credits	58 courses + 16 R&D + thesis	16-24 courses + extensive thesis
Exit Requirement	R&D projects + optional publications	Original thesis + defense
Funding	TA/RAP stipends	Fellowships, projects (higher avg)
Career Path	R&D jobs or PhD bridge	Academia/industry research leads

MS offers flexibility for uncertain PhD commitment, while PhD suits those ready for rigorous, multi-year innovation.[2]

[1](https://acad.iitb.ac.in/admissions/masters/msbyresearch)
[2](https://www.reddit.com/r/GATEtard/comments/1kr2rue/got_offers_in_iit_bombay_cminds_ms_and_iisc_mtech/)
[3](https://acad.iitb.ac.in/files/MS%20by%20Research%20programme.pdf)
[4](https://www.youtube.com/watch?v=I1ZbnA9s-dc)
[5](https://www.reddit.com/r/Indian_Academia/comments/lyzrb2/what_are_the_pros_and_cons_of_doing_an_ms_by/)
[6](https://www.cse.iitb.ac.in/admissions/pg/ms)
[7](https://www.cse.iitb.ac.in/admissions/pg)
[8](https://www.reddit.com/r/GATEtard/comments/1bqsftf/is_ms_research_in_computer_science_or_ai_worth/)
[9](https://www.facebook.com/groups/1015191011986167/posts/1862164757288784/)
[10](https://www.youtube.com/watch?v=oX3xhE1EDzU)

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IIT Bombay MS by Research graduates transitioning to PhD face reduced coursework demands due to credits earned during MS.[3]

Course Credits Needed
PhD requires 16-24 total course credits overall, but MS completers (with 58+ credits already) typically need 0-6 additional credits, often just electives or qualifiers if gaps exist. CPI of 7.5+ from MS semesters qualifies for direct conversion post-Semester 2.[2][3]

Transition Process
Conversion to PhD (or dual-degree path) credits prior MS coursework (e.g., electives, seminars count toward PhD bucket). Focus shifts to thesis, with PhD qualifiers emphasizing research over new courses.[4][2][3]

Comparison Table

Requirement	Full PhD Entry	Post-MS PhD [3]
Total Courses	16-24 credits	0-6 additional
Qualifier CPI	7.5 minimum	Carried from MS
Timeline	4-6 years	2-4 years post-MS

This pathway minimizes redundant coursework for MS alumni.[2]

[1](https://www.iiitb.ac.in/courses/master-of-science-by-researchdoctor-of-philosophy)
[2](https://acad.iitb.ac.in/files/RulesforConversionfromMasterstoDualDegree_0.pdf)
[3](https://acad.iitb.ac.in/files/MS%20by%20Research%20programme.pdf)
[4](https://acad.iitb.ac.in/admissions/research)
[5](https://collegevidya.com/blog/iit-phd-eligibility-minium-qualifications/)
[6](https://www.iitb.ac.in/newacadhome/phdRules.pdf)
[7](https://www.som.iitb.ac.in/doctoral-programme-eligibility/)
[8](https://www.youtube.com/watch?v=UkGgipQTTrE)
[9](https://www.cse.iitb.ac.in/admissions/pg/ms)
[10](https://acad.iitb.ac.in/academics/rules/pg)

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IIT Bombay (often referred to as IIT Mumbai) offers an MS by Research program, which serves as a pathway to PhD studies, with specific course credit requirements tailored to the student's prior qualifications. For PhD admission after MS by Research, candidates typically need to meet minimum credit thresholds from their MS phase, focusing on research credits alongside limited coursework.

MS by Research Credits
The MS by Research program requires a total minimum of around 58 course credits plus 6 credits for communication skills, divided into theory, labs, seminars, and R&D projects. This includes core areas like computing (6 credits), probability/statistics (6 credits), seminars (4 credits), and electives, spread over 4-6 semesters with decreasing credit loads (e.g., 18-32 in semester 1, down to 0-12 in later semesters).[4][5]

PhD Course Requirements
PhD students entering with an MS (like MS by Research) or equivalent need a minimum of 16 credits, consisting of 2 courses plus a seminar, up to a maximum of 22 credits; these can be from any IIT Bombay academic unit. Other entry backgrounds require higher loads (e.g., 40-64 credits for M.Mgt. holders or 52-64 for others), but MS holders benefit from reduced coursework to prioritize thesis research.[2]

Transition Pathway
The MS by Research is explicitly designed as a bridge to PhD, often lasting 18 months to 3 years, with no fixed course plan but emphasis on R&D projects (e.g., 16 credits total in MS R&D 1 and 2) and seminars; PhD continuation builds directly on this with the minimal 16-credit top-up. Residency for PhD is typically 10-12 semesters maximum when combined with prior degrees.[1][5]

[1](https://www.iiit.ac.in/academic-regulations-for-ms-by-research-programmes/)
[2](https://www.iitb.ac.in/newacadhome/phdRules.pdf)
[3](https://iimmumbai.ac.in/regular-phd-programme-eligibility-criteria)
[4](https://acad.iitb.ac.in/files/MS%20by%20Research%20programme.pdf)
[5](https://acad.iitb.ac.in/admissions/masters/msbyresearch)
[6](https://www.scribd.com/document/792963893/MAbyResearchprogramme)
[7](https://www.shiksha.com/university/iit-bombay-indian-institute-of-technology-mumbai-54212/courses/phd-bc)
[8](https://www.facebook.com/groups/1015191011986167/posts/2278571588981430/)
[9](https://acad.iitb.ac.in/admissions/research)
[10](https://collegevidya.com/university/iit-bombay/part-time-phd/)
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At IIT Bombay's Department of Computer Science and Engineering (CSE), PhD students entering with an MS by Research complete the minimum 16 credits (2 courses plus a seminar, up to 22 credits maximum) using advanced CSE offerings. These are selected in consultation with the advisor to align with the thesis focus, emphasizing depth over breadth.

Core CSE Options
Eligible subjects include advanced theory like Algorithms and Complexity (e.g., CS 651: Advanced Algorithms), Systems (e.g., CS 663: Operating Systems or CS 675: Computer Networks), and Programming Languages (e.g., CS 641: Compilers). Seminars such as CS 690 (Research Trends Seminar) fulfill the seminar requirement, often involving presentations on current CSE research areas like AI, Machine Learning, or Data Science.[1][3][5]

Elective Flexibility
Electives can draw from interdisciplinary CSE areas like CS 748: Machine Learning, CS 663: Distributed Systems, or CS 709: Formal Methods, available across CSE and related centers; credits are typically 6 per course (12 for two) plus 4 for seminar. No fixed list applies—choices prioritize relevance to the student's MS research continuity, with completion expected early in the PhD for swift thesis progression.[3][5][1]

[1](https://www.shiksha.com/university/iit-bombay-indian-institute-of-technology-mumbai-54212/course-m-tech-ph-d-in-computer-science-and-engineering-1104363)
[2](https://catalog.iit.edu/graduate/colleges/computing/computer-science/phd-computer-science/phd-computer-science.pdf)
[3](https://www.iitb.ac.in/newacadhome/phdRules.pdf)
[4](https://zollege.in/university/185024-iit-bombay-indian-institute-of-technology-iitb-mumbai/phd-computer-science-and-engineering-2691)
[5](https://www.cse.iitb.ac.in/admissions/pg/phd)
[6](https://www.iitb.ac.in/newacadhome/Ph.D.Brochure2023-24.pdf)
[7](https://www.cse.iitb.ac.in/admissions/pg)
[8](https://oia.osu.edu/news/2025/09/29/ohio-state-iit-bombay-launch-new-dual-degree-phd-program)
[9](https://acad.iitb.ac.in/admissions/research)
[10](https://www.iitb.ac.in/academichome/Ph.D.Brochure2023-24.pdf)
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